



实证资产定价:Momentum

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本节授课内容

- Jegadeesh and Titman 1993 JF
- 动量效应复制编程代码
- 推荐阅读：动量效应后续研究
 - 动量效应的解释
 - 动量效应在中国

动量效应

- Jegadeesh and Titman 1993
- 在过去3~12个月表现最佳（最差）的股票，在接下来3~12个月有延续原来运动方向的趋势

论文信息



Article

Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency

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First published: March 1993 | <https://doi.org/10.1111/j.1540-6261.1993.tb04702.x> | Citations: 6,172

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ABSTRACT

This paper documents that strategies which buy stocks that have performed well in the past and sell stocks that have performed poorly in the past generate significant positive returns over 3-to 12-month holding periods. We find that the profitability of these strategies are not due to their systematic risk or to delayed stock price reactions to common factors. However, part of the abnormal returns generated in the first year after portfolio formation dissipates in the following two years. A similar pattern of returns around the earnings announcements of past winners and losers is also documented.



Volume 48, Issue 1

March 1993

Pages 65-91



References



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动量效应研究背景

- 1993年的金融业界
- Relative strength strategies: buy past winners and sell past losers
- Relative strength是金融实务者的选股标准之一
 - 公募基金倾向于购买过去一个季度价格上升的股票
 - Value Line rankings
 - Relative strength策略能产生超额收益

动量效应研究背景

- 1993年的金融学术界
- Contrarian strategies: buying past losers and selling past winners
- De Bondt and Thaler (1985, 1987)
 - Stock prices overreact to information
 - Stocks that performed poorly over the previous **3 to 5 years** achieve higher returns than stocks that performed well over the same period.
 - But, interpretations are mixed, and only hold on January

动量效应研究背景

- 1993年的金融学术界
- Jegadeesh (1990) and Lehmann (1990)
 - Short-term reversal: contrarian strategies that select stocks based on their returns in the **previous week or month** generate significant abnormal returns
 - But, may reflect the short-term price pressure, or lack of liquidity in the market

思考

- 业界和学术界的不一致发现如何解释？
- 假说1：实务者的超额收益是spurious
- 假说2：实务者和学术界研究的time horizons不同
 - Short-term return reversals: 1 week or 1 month
 - Long-term return reversals: 3 to 5 years
 - Past 3 to 12 months

动量效应研究内容

- 研究样本：1965-1989 NYSE/AMEX股票
- 主要研究发现：
 - Pattern：买入过去3-12月表现好股票，卖出过去3-12月表现差股票，能产生未来正向收益
 - Interpretation：not systematic risk or delayed stock price reactions to common factors, but the delayed price reactions to firm-specific info
 - Experience negative abnormal returns after 12 months holding period
 - Similar in returns around earnings announcement

I. 交易策略

- 构建期：基于股票过去1, 2, 3, 4季度的收益率
- 持有期：未来持有投资组合1, 2, 3, 4季度
- $4*4 = 16$ 个策略
 - 版本2：skip a week between the portfolio formation period and holding period to avoid some of the bid-ask spread, price pressure, and lagged reaction effects

I. 交易策略

- 增强策略的检验效果
- In any given month t , the strategies hold a series of portfolios that are selected in the current month as well as in the previous $K-1$ months, where K is the holding period.

I. 交易策略

- **J-month/K-month strategy**
- Select stocks on the basis of returns over the past J months and holds them for K months
- 1. At the beginning of each month t , the securities are ranked in ascending order on the basis of their returns in the past J months.
- 2. Based on these rankings, **ten decile** portfolios are formed that **equally weight** the stocks contained in the top decile, the second decile, and so on. The top decile portfolio is called the “**losers**” decile and the bottom decile is called the “**winners**” decile.

I. 交易策略

- **J-month/K-month strategy**
- 3. In each month t , the strategy buys the winner portfolio and sells the loser portfolio, holding this position for K months
- 4. In addition, the strategy closes out the position initiated in month $t-K$. Hence, under this strategy we revise the weights on $\frac{1}{K}$ of the securities in the entire portfolio in any given month and carry over the rest of the previous month.
- Data: 1965 to 1989 from CRSP

II. 策略收益

Table I

Returns of Relative Strength Portfolios

The relative strength portfolios are formed based on J -month lagged returns and held for K months. The values of J and K for the different strategies are indicated in the first column and row, respectively. The stocks are ranked in ascending order on the basis of J -month lagged returns and an equally weighted portfolio of stocks in the lowest past return decile is the *sell* portfolio and an equally weighted portfolio of the stocks in the highest return decile is the *buy* portfolio. The average monthly returns of these portfolios are presented in this table. The relative strength portfolios in Panel A are formed immediately after the lagged returns are measured for the purpose of portfolio formation. The relative strength portfolios in Panel B are formed 1 week after the lagged returns used for forming these portfolios are measured. The t -statistics are reported in parentheses. The sample period is January 1965 to December 1989.

II. 策略收益

		Panel A				Panel B					
J		$K =$	3	6	9	12	$K =$	3	6	9	12
3	Sell		0.0108 (2.16)	0.0091 (1.87)	0.0092 (1.92)	0.0087 (1.87)		0.0083 (1.67)	0.0079 (1.64)	0.0084 (1.77)	0.0083 (1.79)
3	Buy		0.0140 (3.57)	0.0149 (3.78)	0.0152 (3.83)	0.0156 (3.89)		0.0156 (3.95)	0.0158 (3.98)	0.0158 (3.96)	0.0160 (3.98)
3	Buy-sell		0.0032 (1.10)	0.0058 (2.29)	0.0061 (2.69)	0.0069 (3.53)		0.0073 (2.61)	0.0078 (3.16)	0.0074 (3.36)	0.0077 (4.00)
6	Sell		0.0087 (1.67)	0.0079 (1.56)	0.0072 (1.48)	0.0080 (1.66)		0.0066 (1.28)	0.0068 (1.35)	0.0067 (1.38)	0.0076 (1.58)
6	Buy		0.0171 (4.28)	0.0174 (4.33)	0.0174 (4.31)	0.0166 (4.13)		0.0179 (4.47)	0.0178 (4.41)	0.0175 (4.32)	0.0166 (4.13)
6	Buy-sell		0.0084 (2.44)	0.0095 (3.07)	0.0102 (3.76)	0.0086 (3.36)		0.0114 (3.37)	0.0110 (3.61)	0.0108 (4.01)	0.0090 (3.54)
9	Sell		0.0077 (1.47)	0.0065 (1.29)	0.0071 (1.43)	0.0082 (1.66)		0.0058 (1.13)	0.0058 (1.15)	0.0066 (1.34)	0.0078 (1.59)
9	Buy		0.0186 (4.56)	0.0186 (4.53)	0.0176 (4.30)	0.0164 (4.03)		0.0193 (4.72)	0.0188 (4.56)	0.0176 (4.30)	0.0164 (4.04)
9	Buy-sell		0.0109 (3.03)	0.0121 (3.78)	0.0105 (3.47)	0.0082 (2.89)		0.0135 (3.85)	0.0130 (4.09)	0.0109 (3.67)	0.0085 (3.04)
12	Sell		0.0060 (1.17)	0.0065 (1.29)	0.0075 (1.48)	0.0087 (1.74)		0.0048 (0.93)	0.0058 (1.15)	0.0070 (1.40)	0.0085 (1.71)
12	Buy		0.0192 (4.63)	0.0179 (4.36)	0.0168 (4.10)	0.0155 (3.81)		0.0196 (4.73)	0.0179 (4.36)	0.0167 (4.09)	0.0154 (3.79)
12	Buy-sell		0.0131 (3.74)	0.0114 (3.40)	0.0093 (2.95)	0.0068 (2.25)		0.0149 (4.28)	0.0121 (3.65)	0.0096 (3.09)	0.0069 (2.31)

II. 策略收益

- 复制代码讲解

III. 策略收益来源

- 分解超额收益率，识别收益来源

$$\begin{aligned} r_{it} &= \mu_i + b_i f_t + e_{it}, \\ E(f_t) &= 0 \\ E(e_{it}) &= 0 \\ \text{Cov}(e_{it}, f_t) &= 0, \quad \forall i \\ \text{Cov}(e_{it}, e_{jt-1}) &= 0, \quad \forall i \neq j \end{aligned} \tag{1}$$

where μ_i is the unconditional expected return on security i , r_{it} is the return on security i , f_t is the unconditional unexpected return on a factor-mimicking portfolio, e_{it} is the firm-specific component of return at time t , and b_i is the factor sensitivity of security i . For the 6-month/6-month strategy that we consider in the rest of this paper the length of a period is 6 months.

III. 策略收益来源

- Momentum: stocks that generate higher than average returns in one period also generate higher than average returns in the period that follows.

$$E(r_{it} - \bar{r}_t | r_{it-1} - \bar{r}_{t-1} > 0) > 0$$

and

$$E(r_{it} - \bar{r}_t | r_{it-1} - \bar{r}_{t-1} < 0) < 0,$$

where a bar above a variable denotes its cross-sectional average.

Therefore,

$$E\{(r_{it} - \bar{r}_t)(r_{it-1} - \bar{r}_{t-1})\} > 0. \quad (2)$$

III. 策略收益来源

- Decomposition:

Given the one-factor model defined in (1), the WRSS profits given in expression (2) can be decomposed into the following three terms:

$$\begin{aligned} E\{(r_{it} - \bar{r}_t)(r_{it-1} - \bar{r}_{t-1})\} &= \sigma_\mu^2 + \sigma_b^2 \text{Cov}(f_t, f_{t-1}) \\ &\quad + \overline{\text{Cov}_i}(e_{it}, e_{it-1}), \end{aligned} \tag{3}$$

where σ_μ^2 and σ_b^2 are the cross-sectional variances of expected returns and factor sensitivities respectively.

III. 策略收益来源

- Three potential sources of the relative strength profits:
- Term 1: Cross-sectional dispersion in expected returns
- Term 2: Potential to time the factor: when conditional expectation of the factor portfolio return is high, pick stock with high b
- Term 3: The average serial covariance of the idiosyncratic components of security returns

III. 策略收益来源

- Term 1: The strategy may systematically pick high-risk stocks, reject

Table II
Betas and Market Capitalization of Relative Strength
Portfolios

The relative strength portfolios are formed based on 6-month lagged returns and held for 6 months. The stocks are ranked in ascending order on the basis of 6-month lagged returns. The equally weighted portfolio of stocks in the lowest past return decile is portfolio P1, the equally weighted portfolio of stocks in the next decile is portfolio P2, and so on. The betas with respect to the value-weighted index and the average market capitalizations of the stocks included in these portfolios are reported here. The sample period is January 1965 to December 1989.

	Beta	Average Market Capitalization
P1	1.36	208.24
P2	1.19	480.07
P3	1.14	545.31
P4	1.11	618.85
P5	1.09	692.89
P6	1.08	702.51
P7	1.09	738.09
P8	1.12	758.87
P9	1.17	680.18
P10	1.28	495.13
P10-P1	-0.08	—

III. 策略收益来源

- Term 2: the in-sample serial covariance of the market return is required to be positive: reject

from our decomposition. Given the model expressed in (1), the serial covariance of an equally weighted portfolio of a large number of stocks is:⁸

$$\text{cov}(\bar{r}_t, \bar{r}_{t-1}) = \bar{b}_i^2 \text{Cov}(f_t, f_{t-1}). \quad (4)$$

- $\text{Cov}(f_t, f_{t-1}) = -0.0028$
- $\text{Cov}(e_{it}, e_{it-1}) = 0.0012$
- May arise from stocks underreacting to firm-specific information (support term 3)

IV. 子样本

Table III
Returns of Size-Based and Beta-Based Relative Strength Portfolios

The relative strength portfolios are formed based on 6-month lagged returns and held for 6 months. The stocks are ranked in ascending order on the basis of 6-month lagged returns and the equally weighted portfolio of stocks in the lowest past return decile is portfolio P1, the equally weighted portfolio of stocks in the next decile is portfolio P2, and so on. Average monthly returns and excess returns of these portfolios and the returns of the relative strength portfolios formed using size-based and beta-based subsamples of securities are reported here. The subsample S1 contains the smallest firms, S2 contains the medium-sized firms, and S3 contains the largest firms. The subsamples β_1 , β_2 , and β_3 contain the firms with the smallest, medium, and the largest Scholes-Williams betas estimated from the returns data in the calendar year prior to portfolio formation. The sample period is January 1965 to December 1989.

Panel A: Average Monthly Returns							
	All	S1	S2	S3	β_1	β_2	β_3
P1	0.0079 (1.56)	0.0083 (1.35)	0.0047 (0.99)	0.0082 (2.22)	0.0129 (2.92)	0.0097 (2.01)	0.0052 (0.95)
P2	0.0112 (2.78)	0.0117 (2.29)	0.0102 (2.54)	0.0098 (3.08)	0.0140 (4.38)	0.0128 (3.37)	0.0086 (1.83)
P3	0.0125 (3.40)	0.0152 (3.23)	0.0125 (3.34)	0.0105 (3.53)	0.0132 (4.59)	0.0133 (3.77)	0.0102 (2.28)
P4	0.0124 (3.59)	0.0163 (3.59)	0.0130 (3.58)	0.0105 (3.66)	0.0134 (5.02)	0.0128 (3.82)	0.0110 (2.50)
P5	0.0128 (3.87)	0.0164 (3.74)	0.0134 (3.83)	0.0109 (3.85)	0.0135 (5.14)	0.0135 (4.15)	0.0121 (2.86)
P6	0.0134 (4.14)	0.0174 (4.08)	0.0146 (4.22)	0.0102 (3.66)	0.0135 (5.23)	0.0142 (4.38)	0.0122 (2.92)
P7	0.0136 (4.19)	0.0175 (4.13)	0.0143 (4.12)	0.0109 (3.90)	0.0136 (5.09)	0.0142 (4.43)	0.0126 (3.01)
P8	0.0143 (4.30)	0.0174 (4.11)	0.0148 (4.16)	0.0111 (3.86)	0.0143 (5.12)	0.0146 (4.44)	0.0132 (3.15)
P9	0.0153 (4.36)	0.0183 (4.28)	0.0154 (4.11)	0.0126 (4.17)	0.0165 (5.34)	0.0156 (4.56)	0.0141 (3.28)
P10	0.0174 (4.33)	0.0182 (3.99)	0.0173 (4.11)	0.0157 (4.41)	0.0191 (5.17)	0.0176 (4.53)	0.0160 (3.50)
P10-P1	0.0095 (3.07)	0.0099 (2.77)	0.0126 (4.57)	0.0075 (3.03)	0.0062 (2.05)	0.0079 (2.64)	0.0108 (3.35)
F-Statistics ^a	2.83	2.65	4.51	4.38	2.51	1.99	1.69
p-Value	(0.00)	(0.00)	(0.00)	(0.00)	(0.01)	(0.04)	(0.09)

V. 子区间

Table IV
Returns on Size-Based Relative Strength Portfolios (P10–P1)
by Calendar Months

The relative strength portfolios are formed based on 6-month lagged returns and held for 6 months. The stocks are ranked in ascending order on the basis of 6-month lagged returns and the equally weighted portfolio of stocks in the lowest past return decile is the *sell* portfolio and the equally weighted portfolio of stocks in the highest past return decile is the *buy* portfolio. This table reports the average monthly returns of the zero-cost, buy minus sell, portfolio in each calendar month. The average returns of the zero-cost portfolios formed using size-based subsamples of securities are also reported. The subsample S1 contains the smallest firms, S2 contains the medium-sized firms, and S3 contains the largest firms. The sample period is January 1965 to December 1989.

	All	S1	S2	S3
Jan.	−0.0686 (−3.52)	−0.0797 (−3.36)	−0.0347 (−2.14)	−0.0161 (−1.28)
Feb.	0.0063 (0.85)	0.0089 (0.81)	0.0149 (2.44)	0.0099 (1.35)
Mar.	0.0105 (1.37)	0.0196 (2.08)	0.0103 (1.49)	0.0108 (1.49)
Apr.	0.0333 (7.39)	0.0323 (5.35)	0.0368 (7.29)	0.0215 (4.91)
May	0.0102 (1.32)	0.0046 (0.56)	0.0091 (1.18)	0.0079 (1.19)
June	0.0238 (3.86)	0.0237 (3.50)	0.0231 (3.23)	0.0185 (2.59)
July	0.0075 (0.96)	0.0112 (1.44)	0.0084 (0.96)	0.0035 (0.41)
Aug.	0.0027 (0.35)	0.0079 (0.97)	−0.0011 (−0.14)	−0.0058 (−0.71)
Sept.	0.0116 (1.10)	0.0126 (1.20)	0.0137 (1.27)	0.0053 (0.60)
Oct.	0.0137 (1.30)	0.0160 (1.40)	0.0151 (1.44)	0.0025 (0.22)
Nov.	0.0372 (5.31)	0.0352 (5.01)	0.0331 (4.12)	0.0248 (2.78)
Dec.	0.0264 (2.61)	0.0265 (2.13)	0.0224 (2.86)	0.0070 (0.99)
Feb.–Dec.	0.0166 (6.67)	0.0181 (6.47)	0.0169 (6.83)	0.0096 (4.00)
<i>F</i> -Statistics ^a	7.90	7.14	4.11	1.81
<i>p</i> -Value	(0.00)	(0.00)	(0.00)	(0.51)
<i>F</i> -Statistics ^b	2.04	1.23	1.91	1.28
<i>p</i> -Value	(0.03)	(0.27)	(0.04)	(0.24)

V. 子区间

Table VI
Returns of Size-Based Relative Strength Portfolios: Subperiod Analysis

The relative strength portfolios are formed based on 6-month lagged returns and held for 6 months. The stocks are ranked in ascending order on the basis of 6-month lagged returns and the equally weighted portfolio of stocks in the lowest past return decile is the *sell* portfolio and the equally weighted portfolio of stocks in the highest past return decile is the *buy* portfolio. This table reports the average monthly returns of the zero-cost, buy minus sell, portfolio within 5-year subperiods. The average returns of the zero-cost portfolios formed using size-based subsamples of securities within subperiods are also reported. The subsample S1 contains the smallest firms, S2 contains the medium-sized firms, and S3 contains the largest firms. The sample period is January 1965 to December 1989.

Sample	Months	65-69	70-74	75-79	80-84	85-89
All	All	0.0123 (1.94)	0.0109 (1.23)	-0.0044 (-0.51)	0.0127 (2.67)	0.0162 (3.42)
	Jan.	-0.0524 (-1.28)	-0.1070 (-2.54)	-0.1017 (-1.31)	-0.0253 (-1.38)	-0.0569 (-2.76)
	Feb.-Dec.	0.0182 (3.36)	0.0217 (2.88)	0.0044 (0.78)	0.0161 (3.44)	0.0229 (6.09)
S1	All	0.0082 (1.14)	0.0128 (1.63)	-0.0064 (-0.58)	0.0153 (2.61)	0.0197 (2.89)
	Jan.	-0.0838 (-1.60)	-0.0853 (-2.29)	-0.1107 (-1.09)	-0.0124 (-0.62)	-0.1064 (-4.45)
	Feb.-Dec.	0.0165 (3.19)	0.0217 (3.18)	0.0031 (0.41)	0.0179 (2.94)	0.0311 (6.59)
S2	All	0.0177 (3.08)	0.0115 (1.57)	0.0018 (0.24)	0.0172 (3.38)	0.0146 (3.40)
	Jan.	-0.0264 (-1.05)	-0.0465 (-1.81)	-0.0795 (-1.16)	-0.0100 (-0.46)	-0.0112 (-0.48)
	Feb.-Dec.	0.0217 (3.86)	0.0168 (2.29)	0.0092 (1.87)	0.0197 (3.83)	0.0170 (4.08)
S3	All	0.0129 (2.71)	0.0115 (1.62)	0.0018 (0.35)	0.0076 (1.41)	0.0035 (0.73)
	Jan.	-0.0073 (-0.32)	-0.0154 (-0.48)	-0.0335 (-0.77)	-0.0094 (-0.33)	-0.0147 (-0.78)
	Feb.-Dec.	0.0148 (3.08)	0.0139 (1.95)	0.0050 (1.21)	0.0092 (1.70)	0.0052 (1.04)

V. 子区

Table VII

Performance of Relative Strength Portfolios in Event Time

The relative strength portfolios are formed based on 6-month lagged returns. The stocks are ranked in ascending order on the basis of 6-month lagged returns. The equally weighted portfolio of stocks in the lowest past return decile is the *sell* portfolio and the equally weighted portfolio of stocks in the highest past return decile is the *buy* portfolio. This table reports the average returns of the zero-cost, buy minus sell, portfolio in each month following the formation period. t is the month after portfolio formation. The sample period is January 1965 to December 1989. Autocorrelation-consistent estimates of standard errors are used to compute the t -statistics for cumulative returns.

t	Monthly Return	Cumulative Return	t	Monthly Return	Cumulative Return	t	Monthly Return	Cumulative Return
1	-0.0025 (-0.59)	-0.0025 (-0.59)	13	-0.0036 (-1.12)	0.0915 (3.35)	25	-0.0035 (-1.36)	0.0521 (1.41)
2	0.0124 (3.29)	0.0099 (1.37)	14	-0.0039 (-1.34)	0.0876 (3.07)	26	-0.0030 (-1.14)	0.0492 (1.22)
3	0.0116 (3.18)	0.0216 (2.20)	15	-0.0034 (-1.21)	0.0842 (2.89)	27	-0.0024 (-0.98)	0.0467 (1.10)
4	0.0110 (3.19)	0.0326 (2.67)	16	-0.0038 (-1.41)	0.0804 (2.76)	28	-0.0032 (-1.33)	0.0435 (0.98)
5	0.0093 (2.82)	0.0419 (2.79)	17	-0.0047 (-1.74)	0.0757 (2.70)	29	-0.0032 (-1.38)	0.0403 (0.87)
6	0.0091 (2.94)	0.0510 (2.92)	18	-0.0056 (-2.19)	0.0701 (2.68)	30	-0.0030 (-1.31)	0.0373 (0.77)
7	0.0134 (4.98)	0.0644 (3.32)	19	-0.0026 (-1.14)	0.0675 (2.75)	31	-0.0001 (-0.06)	0.0372 (0.74)
8	0.0115 (4.16)	0.0759 (3.60)	20	-0.0032 (-1.35)	0.0642 (2.73)	32	0.0008 (0.41)	0.0380 (0.73)
9	0.0085 (3.07)	0.0844 (3.73)	21	-0.0032 (-1.32)	0.0611 (2.55)	33	0.0013 (0.62)	0.0394 (0.73)
10	0.0048 (1.69)	0.0892 (3.74)	22	-0.0034 (-1.39)	0.0577 (2.21)	34	0.0008 (0.36)	0.0402 (0.71)
11	0.0045 (1.55)	0.0938 (3.77)	23	-0.0011 (-0.45)	0.0566 (1.93)	35	0.0010 (0.45)	0.0412 (0.71)
12	0.0013 (0.43)	0.0951 (3.67)	24	-0.0010 (-0.40)	0.0556 (1.69)	36	-0.0005 (-0.24)	0.0406 (0.67)

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- Greater volatility
- Market mean reversion

Table VIII
Back-Testing the Strategy: Performance of Relative Strength
Portfolios Prior to 1965

The relative strength portfolios are formed based on 6-month lagged returns. The stocks are ranked in ascending order on the basis of 6-month lagged returns. The equally weighted portfolio of stocks in the lowest past return decile is the *sell* portfolio and the equally weighted portfolio of stocks in the highest past return decile is the *buy* portfolio. This table reports the average returns of the zero-cost, buy minus sell, portfolio in each month following the formation period. t is the month after portfolio formation. Autocorrelation consistent estimates of standard errors are used to compute the t -statistics for cumulative returns.

Panel A: 1927–1940								
t	Monthly Return	Cumulative Return	t	Monthly Return	Cumulative Return	t	Monthly Return	Cumulative Return
1	−0.0495 (−3.72)	−0.0495 (−3.72)	13	−0.0245 (−2.60)	−0.1257 (−1.50)	25	−0.0118 (−1.41)	−0.3359 (−2.48)
2	−0.0143 (−1.32)	−0.0639 (−2.21)	14	−0.0166 (−2.08)	−0.1423 (−1.69)	26	−0.0067 (−1.01)	−0.3427 (−2.53)
3	−0.0088 (−0.87)	−0.0726 (−1.78)	15	−0.0164 (−1.87)	−0.1587 (−1.83)	27	−0.0135 (−1.82)	−0.3562 (−2.52)
4	−0.0048 (−0.45)	−0.0775 (−1.60)	16	−0.0200 (−2.20)	−0.1787 (−2.01)	28	−0.0082 (−1.06)	−0.3644 (−2.47)
5	0.0061 (0.60)	−0.0713 (−1.40)	17	−0.0131 (−1.80)	−0.1919 (−2.12)	29	−0.0125 (−1.37)	−0.3769 (−2.39)
6	0.0057 (0.55)	−0.0656 (−1.22)	18	−0.0166 (−2.11)	−0.2085 (−2.07)	30	−0.0107 (−1.20)	−0.3876 (−2.29)
7	0.0092 (0.83)	−0.0564 (−1.05)	19	−0.0161 (−1.90)	−0.2245 (−2.01)	31	−0.0018 (−0.20)	−0.3894 (−2.18)
8	0.0054 (0.52)	−0.0511 (−0.92)	20	−0.0224 (−2.28)	−0.2469 (−2.03)	32	−0.0022 (−0.26)	−0.3916 (−2.07)
9	−0.0029 (−0.34)	−0.0539 (−0.94)	21	−0.0178 (−1.92)	−0.2647 (−2.04)	33	0.0008 (0.11)	−0.3908 (−1.99)
10	−0.0065 (−0.68)	−0.0604 (−0.90)	22	−0.0213 (−2.08)	−0.2860 (−2.14)	34	−0.0025 (−0.41)	−0.3933 (−1.97)
11	−0.0183 (−1.74)	−0.0787 (−1.04)	23	−0.0183 (−1.74)	−0.3043 (−2.23)	35	−0.0050 (−0.89)	−0.3983 (−1.97)
12	−0.0225 (−2.35)	−0.1012 (−1.27)	24	−0.0198 (−1.94)	−0.3241 (−2.41)	36	−0.0098 (−1.47)	−0.4081 (−2.01)

VII 回測

Table VIII—Continued

Panel B: 1941–1964								
<i>t</i>	Monthly Return	Cumulative Return	<i>t</i>	Monthly Return	Cumulative Return	<i>t</i>	Monthly Return	Cumulative Return
1	−0.0035 (−1.04)	−0.0035 (−1.04)	13	−0.0068 (−2.14)	0.0515 (2.57)	25	−0.0035 (−1.32)	0.0014 (0.04)
2	0.0069 (2.32)	0.0034 (0.59)	14	−0.0085 (−3.07)	0.0429 (1.90)	26	−0.0027 (−1.08)	−0.0013 (−0.03)
3	0.0109 (4.15)	0.0143 (2.20)	15	−0.0059 (−2.40)	0.0371 (1.54)	27	−0.0015 (−0.69)	−0.0028 (−0.07)
4	0.0098 (3.81)	0.0241 (3.15)	16	−0.0063 (−2.80)	0.0308 (1.21)	28	−0.0003 (−0.14)	−0.0030 (−0.08)
5	0.0075 (3.09)	0.0316 (3.40)	17	−0.0080 (−3.70)	0.0228 (0.86)	29	−0.0009 (−0.51)	−0.0039 (−0.11)
6	0.0049 (1.97)	0.0365 (3.42)	18	−0.0074 (−3.63)	0.0153 (0.56)	30	−0.0001 (−0.03)	−0.0040 (−0.12)
7	0.0079 (3.24)	0.0444 (3.82)	19	−0.0033 (−1.61)	0.0120 (0.43)	31	0.0017 (0.98)	−0.0023 (−0.08)
8	0.0062 (2.52)	0.0507 (4.00)	20	−0.0012 (−0.61)	0.0108 (0.38)	32	0.0011 (0.69)	−0.0012 (−0.05)
9	0.0039 (1.63)	0.0546 (3.91)	21	−0.0016 (−0.81)	0.0092 (0.31)	33	−0.0005 (−0.32)	−0.0017 (−0.10)
10	0.0022 (0.96)	0.0568 (3.73)	22	−0.0021 (−1.04)	0.0071 (0.22)	34	−0.0006 (−0.37)	−0.0023 (−0.17)
11	0.0024 (1.00)	0.0592 (3.70)	23	−0.0008 (−0.35)	0.0063 (0.19)	35	−0.0004 (−0.24)	−0.0027 (−0.20)
12	−0.0009 (−0.34)	0.0583 (3.40)	24	−0.0014 (−0.60)	0.0050 (0.14)	36	−0.0004 (−0.28)	−0.0030 (−0.20)

VI. 事件分析

Table IX

Quarterly Earnings Announcement Date Returns

The stocks are ranked in ascending order on the basis of 6-month lagged returns. The stocks in the lowest past return decile are called the *losers* group and the stocks in the highest past return decile is called the *winners* group. The differences between the 3-day returns (returns on days -2 to 0) around quarterly earnings announcements for stocks in the winners group and the losers group are reported here ($r_t^w - r_t^l$). t is the month after the ranking date. The sample period is January 1980 to December 1989.

t	$r_t^w - r_t^l$	t	$r_t^w - r_t^l$	t	$r_t^w - r_t^l$
1	0.0055 (2.75)	13	-0.0055 (-2.56)	25	-0.0002 (-0.11)
2	0.0082 (4.41)	14	-0.0080 (-3.89)	26	-0.0021 (-1.02)
3	0.0082 (4.36)	15	-0.0071 (-4.04)	27	-0.0032 (-1.68)
4	0.0090 (4.88)	16	-0.0097 (-5.75)	28	-0.0028 (-1.31)
5	0.0059 (3.16)	17	-0.0062 (-2.90)	29	-0.0015 (-0.62)
6	0.0058 (3.14)	18	-0.0060 (-2.96)	30	-0.0021 (-1.10)
7	0.0013 (0.62)	19	-0.0031 (-1.63)	31	-0.0027 (-1.52)
8	0.0000 (-0.02)	20	-0.0017 (-0.82)	32	-0.0021 (-1.13)
9	-0.0020 (-1.07)	21	0.0006 (0.27)	33	-0.0020 (-1.05)
10	-0.0031 (-1.60)	22	-0.0005 (-0.29)	34	-0.0017 (-0.91)
11	-0.0039 (-2.23)	23	-0.0001 (-0.05)	35	-0.0022 (-1.29)
12	-0.0053 (-2.75)	24	0.0012 (0.63)	36	-0.0059 (-2.91)

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